

(Vector spaces) (Chapter No. 6)

Vector Space

Let F be a field & V a non empty set on which an operation of addition is defined.

Suppose for every $a \in F$ & $v \in V$, av is an element of V then V is called a vector space over F if the following conditions are satisfied.

(i) V is an abelian gr under addition

(ii) $a(bu) = (ab)u$ $a, b \in F, u \in V$

(iii) $(a+b)u = au + bu$ $a, b \in F, u \in V$

(iv) $a(u+v) = au + av$ $a \in F, u, v \in V$

(v) $1v = v$ where 1 is unity of F

Note (i) The elements of F are called scalars & the elements of V are called vectors

(ii) If V is a vector space over F , we write it as

$$V(F)$$

(iii) We do not multiply elements of V with each other & then we multiply an element of F by an element of V

Example Let V be a vector space over F then

$$(i) a0 = 0 \quad \text{for } a \in F$$

$$(ii) 0v = 0 \quad \text{for } v \in V$$

$$(iii) (-a)v = a(-v) = -av \quad \text{for } a \in F, v \in V$$

$$(iv) \text{ If } av = 0 \text{ then either } a = 0 \text{ or } v = 0$$

$$(v) a(u-v) = au - av \quad \text{for } a \in F, u, v \in V$$

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Example

$$\log_{10} \lambda = 63.279 \text{ E}$$



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Proof

$$1 \cdot 0 = 0 + 0$$

($\therefore 0$ is additive identity of V)

$$\therefore a \cdot 0 = a(0 + 0)$$

$$= a \cdot 0 + a \cdot 0$$

$$a \cdot 0 = a \cdot 0 + a \cdot 0$$

(by iv)

$$0 + a \cdot 0 = a \cdot 0 + a \cdot 0$$

$\therefore 0$ is identity of V

$$0 = a \cdot 0$$

$\therefore V$ is a gr. s. cancellative law holds

$$\therefore \boxed{a \cdot 0 = 0}$$

$$(ii) \quad 0 \cdot v = 0 + 0 \cdot v$$

$\therefore 0$ is additive identity of F

$$0 \cdot v = (0 + 0) \cdot v$$

$$= 0 \cdot v + 0 \cdot v$$

$$\therefore 0 \cdot v = 0 \cdot v + 0 \cdot v$$

$$0 \cdot v + 0 = 0 \cdot v + 0 \cdot v$$

($\therefore 0$ is identity of V)

$$\therefore 0 = 0 \cdot v$$

($\therefore V$ is a gr. s. cancellative law holds)

$$\therefore \boxed{0 \cdot v = 0}$$

$$\begin{aligned} (iii) \quad (-a)v + av &= (-a + a)v \\ &= 0v \\ &= 0 \end{aligned}$$

$\therefore (-a)v + av$ are additive inverses of each other

$$\therefore \boxed{(-a)v = -av}$$

Also

$$a(-v) + av = a(-v - v)$$

by (iv)

$$= a \cdot 0$$

$$= 0$$

$\therefore a(-v) + av$ are additive inverses of each other

Now

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$$\alpha(-u) = -\alpha u$$

$$\text{So } (-\alpha)u = \alpha(-u) = -\alpha u$$

(iv)

Let $a \neq 0$ so that a^{-1} exists s.t. $aa^{-1} = a^{-1}a = 1$

Now

$$0 = 1 \cdot 0 \quad (i, v)$$

$$= (a^{-1}a)0$$

$$= a^{-1}(a0)$$

(by (i))

$$= a^{-1}(0)$$

$$= 0$$

So if $a \neq 0$ then $u = 0$

Similarly if $u \neq 0$ then $a = 0$

So $au = 0 \Rightarrow$ either $a = 0$ or $u = 0$

$$(v) \alpha(u-u) = \alpha(u+(-u))$$

$$= \alpha u + \alpha(-u)$$

by (iv)

$$= \alpha u + (-\alpha u)$$

just proved

$$= \alpha u - \alpha u$$

$$\text{So } \alpha(u-u) = \alpha u - \alpha u$$

Subspace

A non empty subset W of a vector space V is called a subspace of V if W itself is a vector space over F under the same operation as defined in V

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V . Therefore W is a subspace of V

Corollary A non empty subset W of a vector space V is a subspace of V iff for $w_1, w_2 \in W$ & $a, b \in F \Rightarrow aw_1 + bw_2 \in W$

Proof:

We shall establish that Conditions (i) & (ii) of above theorem imply the Condition of this Corollary & viceversa

For this suppose that W is a subspace of V then by (ii) Condition

$$aw_1, bw_2 \in W$$

& therefore by (i) Condition $aw_1 + bw_2 \in W$

Conversely

Let W be a non empty subset of V in which the Condition

$$w_1, w_2 \in W \text{ & } a, b \in F \Rightarrow aw_1 + bw_2 \in W \text{ holds}$$

We shall prove that W is a subspace of V

Take $a=b=1$ in above Condition

$$1.w_1 + 1.w_2 = w_1 + w_2 \in W$$

Now take $b=0$

$$\text{So } aw_1 + 0.w_2 = aw_1 \in W$$

Hence by previous theorem

W is a subspace of V

Thm Let U & W be subspaces of a vector space V over F then $U \cap W$ is also a

Proof

Let $a, b \in F$ & $w_1, w_2 \in U \cap W$

$\Rightarrow w_1, w_2 \in U$ & $w_1, w_2 \in W$

But U & W are subspaces of V

So $aw_1 + bw_2 \in U$

& $aw_1 + bw_2 \in W$

Hence $aw_1 + bw_2 \in U \cap W$

So $a, b \in F$ & $aw_1 + bw_2 \in U \cap W \Rightarrow aw_1 + bw_2 \in U \cap W$

Hence $U \cap W$ is a subspace of V .

Thm The intersection of any no. of subspaces of a vector space V is a subspace of V

Proof

Let $\{U_d : d \in I\}$ be any subcollection of subspaces of a vector space V over the field F . Then we have to prove that $\bigcap_{d \in I} U_d$ is also a subspace of V .

For this let $a, b \in F$ & $u_1, u_2 \in \bigcap_{d \in I} U_d$

$\Rightarrow u_1, u_2 \in U_d$ for all $d \in I$

But each U_d is a subspace of V

So $au_1 + bu_2 \in U_d$ for each $d \in I$

$$\Rightarrow au_1 + bu_2 \in \bigcap_{a \in I} U_2$$

So $\bigcap_{a \in I} U_2$ is a subspace of V .

Sum of two Subspaces

Let U & W be two subspaces of a vector space V . We define $U+W$ as

$$U+W = \{u+w \mid u \in U + w \in W\}$$

Thm 1 If U, W are subspaces of a vector space V then $U+W$ is a subspace of V containing both U, W . Further $U+W$ is the smallest subspace of V containing both U, W .

Proof Since U, W are subspaces of V then we define

$$U+W = \{u+w \mid u \in U + w \in W\}$$

we will prove that $U+W$ is a subspace of V

For this let $a, b \in F$ & $u_1, u_2 \in U+W$

$$\Rightarrow u_1 = u + w_1$$

$$+ u_2 = u_2 + w_2$$

$$\text{where } u_1, u_2 \in U + w_1, w_2 \in W$$

Now since U is a subspace of V so $au_1 + bu_2 \in U$

+ since W is a subspace of V , so $aw_1 + bw_2 \in W$

$$\text{Hence } (au_1 + bu_2) + (aw_1 + bw_2) \in U+W$$

$$\text{or } (au_1 + aw_1) + (bu_2 + bw_2) \in U+W$$

$$a(u_1v_1) + b(u_2v_2) \in U+W$$

$$\text{or } ax_1 + bx_2 \in U+W$$

So for $a, b \in F$ & $x_1, x_2 \in U+W \Rightarrow ax_1 + bx_2 \in U+W$

Hence $U+W$ is a subspace of V

Next we prove that $U+W$ is a subspace of V containing both U & W . i.e., $U \subseteq U+W$ & $W \subseteq U+W$

$$\text{Since } u \in U + 0 \in W$$

$$\Rightarrow u+0 = u \in U+W \quad \text{for all } u \in U$$

$$\text{So } U \subseteq U+W$$

$$\text{Similarly } W \subseteq U+W$$

Hence $U+W$ is a subspace of V containing both U & W

Now we will prove that $U+W$ is the smallest subspace of V containing both U & W

Let S be any subspace of V containing both U & W then for every $u \in U$ & $w \in W$,

We have $u \in S$ & $w \in S$ so that $u+w \in S$

$$\text{But } u+w \in U+W$$

$$\text{So } U+W \subseteq S$$

Hence $U+W$ is the smallest subspace of V containing both U & W .

A vector space V is called the direct sum of its subspaces U & W if

$$(i) V = U + W$$

$$(ii) U \cap W = \{0\}$$

Linear combinations:

Let V be a vector space over a field F & let

$$u_1, u_2, u_3, \dots, u_n \in V$$

Any vector in V of the form

$$a_1 u_1 + a_2 u_2 + \dots + a_n u_n \quad \text{where } a_i \in F$$

is called a linear combination of $u_1, u_2, \dots, u_n$

Linear span:

Let S be a non empty subset of a vector space V Then the set of all linear combinations of finite no. of elements of S is called the linear span of S & is denoted by $\langle S \rangle$

Note: $\langle S \rangle$ is said to be spanned or generated by S & S is called a spanning set for $\langle S \rangle$

Theorem: Let S be a non empty set of vectors in a vector space V over a field F Then $\langle S \rangle$ is a subspace of V containing S & it is the smallest subspace of V containing S

Proof

Let $a, b \in F$ & $u, v \in \langle S \rangle$ then

$$u = a_1 u_1 + a_2 u_2 + \dots + a_n u_n = \sum_{i=1}^n a_i u_i$$

$$v = b_1 u_1 + b_2 u_2 + \dots + b_m u_m = \sum_{j=1}^m b_j u_j$$

where $u_i, u_j \in S$ & $a_i, b_j \in F$

$$\begin{aligned}
 \text{Now } a.u + b.v &= a \left(\sum_{i=1}^n a_i u_i \right) + b \left(\sum_{j=1}^m b_j v_j \right) \\
 &= \sum_{i=1}^n a(a_i u_i) + \sum_{j=1}^m b(b_j v_j) \\
 &= \sum_{i=1}^n (aa_i) u_i + \sum_{j=1}^m (bb_j) v_j \quad \therefore a(bu) = (ab)u
 \end{aligned}$$

which shows that $au + bv$ is a linear combination of vectors in S . So $au + bv \in \langle S \rangle$

So $a, b \in F, u, v \in \langle S \rangle \Rightarrow au + bv \in \langle S \rangle$

Hence $\langle S \rangle$ is a subspace of V .

(Now)

Since for all $u \in S$

$$u = 1.u \Rightarrow u \in \langle S \rangle$$

$$\text{So } S \subseteq \langle S \rangle$$

Hence $\langle S \rangle$ is a subspace of V containing S .

Now we prove that $\langle S \rangle$ is the smallest subspace of V containing S .

If W is any other subspace of V containing S then it contains all vectors of the form $\sum_{i=1}^n a_i u_i$; where $a_i \in F$ & $u_i \in S$.

$$\Rightarrow \langle S \rangle \subseteq W$$

Thus $\langle S \rangle$ is the smallest subspace of V containing S .

Theorem: If S, T are subsets of V then $S \subseteq T$ implies $\langle S \rangle \subseteq \langle T \rangle$

Proof:

$$\text{Let } S = \{u_1, u_2, \dots, u_k\}$$

$$\& T = \{u_1, u_2, \dots, u_k, u_{k+1}, \dots, u_n\}$$

then obviously

$$S \subseteq T$$

We want to show that $\langle S \rangle \subset \langle T \rangle$. For this

let $v \in \langle S \rangle$ then by definition of $\langle S \rangle$, v is a linear combination of vectors $v_1, v_2, \dots, v_r$ of S i.e., $v = a_1 v_1 + a_2 v_2 + \dots + a_r v_r$

Now v can also be written as

$$v = a_1 v_1 + a_2 v_2 + \dots + a_r v_r + 0v_{r+1} + 0v_{r+2} + \dots + 0v_n$$

which shows that v is a linear combination of vectors of T

Hence $v \in \langle T \rangle$

3. $\langle S \rangle \subset \langle T \rangle$

Finite dimensional vector space:

A vector space V is said to be finite dimensional if there is a finite subset S in V such that $V = \langle S \rangle$

Exercise No. 6.1



Q1 Let V be the set of all infinite sequences in a field F with addition & scalar multiplication defined as below:

For $u = \{a_n\} = a_1, a_2, \dots, a_n, \dots \in V$

$v = \{b_n\} = b_1, b_2, \dots, b_n, \dots \in V$

$$u + v = \{a_n\} + \{b_n\} = a_1 + b_1, a_2 + b_2, \dots, a_n + b_n, \dots$$

$$\& k u = k \{a_n\} = k a_1, k a_2, \dots, k a_n, \dots$$

where $a_n, b_n \& k$ are all in F , $n = 1, 2, 3, \dots$

Show that V is a vector space over F .

Sol.

$$\text{Here } V = \{(a_1, a_2, \dots) \mid a_i \in F\}$$

First we prove that $(V, +)$ is an abelian gr.

(a)

(i) Closure law

$$\text{Let } u = (a_1, a_2, \dots)$$

$$\& \mathcal{U}_2 = (b_1, b_2, \dots)$$

$$\begin{aligned} \Rightarrow \mathcal{U}_1 + \mathcal{U}_2 &= (a_1, a_2, \dots) + (b_1, b_2, \dots) \\ &= (a_1 + b_1, a_2 + b_2, \dots) \in V \end{aligned}$$

(ii) Associative Law

$$\text{Let } \mathcal{U}_1 = (a_1, a_2, \dots)$$

$$\mathcal{U}_2 = (b_1, b_2, \dots)$$

$$\& \mathcal{U}_3 = (c_1, c_2, \dots) \in V$$

$$\text{Then we prove } \mathcal{U}_1 + (\mathcal{U}_2 + \mathcal{U}_3) = (\mathcal{U}_1 + \mathcal{U}_2) + \mathcal{U}_3$$

Now

$$\begin{aligned} \mathcal{U}_1 + (\mathcal{U}_2 + \mathcal{U}_3) &= (a_1, a_2, \dots) + [(b_1, b_2, \dots) + (c_1, c_2, \dots)] \\ &= (a_1, a_2, \dots) + [(b_1 + c_1), (b_2 + c_2), \dots] \\ &= [a_1 + (b_1 + c_1), a_2 + (b_2 + c_2), \dots] \\ &= [(a_1 + b_1) + c_1, (a_2 + b_2) + c_2, \dots] \\ &= [(a_1 + b_1), (a_2 + b_2), \dots] + (c_1, c_2, \dots) \\ &= [(a_1, a_2, \dots) + (b_1, b_2, \dots)] + (c_1, c_2, \dots) \\ &= (\mathcal{U}_1 + \mathcal{U}_2) + \mathcal{U}_3 \end{aligned}$$

(iii) Identity Law

Here $0 = (0, 0, \dots)$ is the additive identity in

V because for any $\mathcal{U} = (a_1, a_2, \dots) \in V$

$$\begin{aligned} \mathcal{U} + 0 &= (a_1, a_2, \dots) + (0, 0, \dots) \\ &= (a_1 + 0, a_2 + 0, \dots) \\ &= (a_1, a_2, \dots) \end{aligned}$$

$$\mathcal{U} + 0 = \mathcal{U}$$

$$\text{Similarly } 0 + \mathcal{U} = \mathcal{U}$$

(iv) Inverse Law

Every $\mathcal{U} = (a_1, a_2, \dots) \in V$ has additive inverse

$$-\mathcal{U} = (-a_1, -a_2, \dots) \in V \text{ because}$$

$$\begin{aligned} \mathcal{U} + (-\mathcal{U}) &= (a_1, a_2, \dots) + (-a_1, -a_2, \dots) \\ &= (a_1 - a_1, a_2 - a_2, \dots) \end{aligned}$$

$$u + (-u) = (0, 0, \dots)$$

$$u + (-u) = 0$$

Similarly $-u + u = 0$

(V) Commutative law

Let $u_1 = (a_1, a_2, \dots)$

& $u_2 = (b_1, b_2, \dots)$

Then we prove $u_1 + u_2 = u_2 + u_1$

Now

$$\begin{aligned} u_1 + u_2 &= (a_1, a_2, \dots) + (b_1, b_2, \dots) \\ &= (a_1 + b_1, a_2 + b_2, \dots) \\ &= (b_1 + a_1, b_2 + a_2, \dots) \\ &= (b_1, b_2, \dots) + (a_1, a_2, \dots) \end{aligned}$$

$$u_1 + u_2 = u_2 + u_1$$

Hence $(V, +)$ is an abelian gr.

(b) Scalar multiplication:

(i) Let $a \in F$ & $u = (a_1, a_2, \dots) \in V$ then

$$\begin{aligned} au &= a(a_1, a_2, \dots) \\ &= (aa_1, aa_2, \dots) \in V \end{aligned}$$

(ii)

Let $a, b \in F$ & $u = (u_1, u_2, \dots) \in V$

then we prove $a(bu) = (ab)u$

$$\begin{aligned} \text{Now } a(bu) &= a[b(a_1, a_2, \dots)] \\ &= a[(ba_1, ba_2, \dots)] \\ &= [a(ba_1), a(ba_2), \dots] \\ &= [(ab)a_1, (ab)a_2, \dots] \\ &= (ab)(a_1, a_2, \dots) \end{aligned}$$

$$\therefore a(bu) = (ab)u$$

(iii)

Let $a, b \in F$ & $u = (a_1, a_2, \dots)$

then we prove $(a+b)v = av + bv$ 519

Now

$$\begin{aligned}(a+b)v &= (a+b)(a_1, a_2, \dots) \\&= [(a+b)a_1, (a+b)a_2, \dots] \\&= [(aa_1 + ba_1), (aa_2 + ba_2), \dots] \\&= (aa_1, aa_2, \dots) + (ba_1, ba_2, \dots) \\&= a(a_1, a_2, \dots) + b(a_1, a_2, \dots)\end{aligned}$$

$$(a+b)v = av + bv$$

(iv) Let $a \in F$ & $v_1 = (a_1, a_2, \dots)$, $v_2 = (b_1, b_2, \dots) \in V$

then we show $a(v_1 + v_2) = av_1 + av_2$

Now

$$\begin{aligned}a(v_1 + v_2) &= a[(a_1, a_2, \dots) + (b_1, b_2, \dots)] \\&= a[(a_1 + b_1, a_2 + b_2, \dots)] \\&= [a(a_1 + b_1), a(a_2 + b_2), \dots] \\&= [(aa_1 + ab_1, aa_2 + ab_2, \dots)] \\&= (aa_1, aa_2, \dots) + (ab_1, ab_2, \dots) \\&= a(a_1, a_2, \dots) + a(b_1, b_2, \dots)\end{aligned}$$

$$a(v_1 + v_2) = av_1 + av_2$$

(v) Let $1 \in F$ & $v = (a_1, a_2, \dots) \in V$ then we prove

$$1 \cdot v = v$$

$$\begin{aligned}\text{Now } 1 \cdot v &= 1 \cdot (a_1, a_2, \dots) \\&= (1 \cdot a_1, 1 \cdot a_2, \dots) \\&= (a_1, a_2, \dots)\end{aligned}$$

$$1 \cdot v = v$$

Since all conditions are satisfied.

S. V is a vector space over F .

Q2 Let V be the set of all ordered pairs of real no's.¹⁵
 Check whether V is a vector space over $\mathbb{R}$ w.r.t.
 the indicated operations. If not, state the axioms
 which fail to hold.

$$(i) (a,b) + (c,d) = (a+c, b+d)$$

$$+ K(a,b) = (Ka, b)$$

Sol. Here $V = \{(a,b) | a, b \in \mathbb{R}\}$

$$+ (a,b) + (c,d) = (a+c, b+d)$$

$$K(a,b) = (Ka, b)$$

First we prove that $(V, +)$ is an abelian gr.

(a)

(i) closure law

$$\text{Let } u_1 = (a,b) \in V$$

$$+ u_2 = (c,d)$$

$$\text{Then } u_1 + u_2 = (a,b) + (c,d) \\ = (a+c, b+d) \in V$$

(ii) Associative law

$$\text{Let } u_1 = (a,b), u_2 = (c,d), u_3 = (e,f) \in V \quad \text{Then}$$

$$\text{we prove } u_1 + (u_2 + u_3) = (u_1 + u_2) + u_3$$

Now

$$\begin{aligned} u_1 + (u_2 + u_3) &= (a,b) + ((c,d) + (e,f)) \\ &= (a,b) + (c+e, d+f) \\ &= (a+(c+e), b+(d+f)) \\ &= ((a+c)+e, (b+d)+f) \\ &= (a+c, b+d) + (e,f) \\ &= ((a,b) + (c,d)) + (e,f) \end{aligned}$$

$$u_1 + (u_2 + u_3) = (u_1 + u_2) + u_3$$

(iii) Identity law

Here $O = (0,0)$ is the additive identity in V

because for $u = (a,b) \in V$

$$O + u = (0,0) + (a,b)$$

$$= (0+a, 0+b)$$

$$= (a,b)$$

$$O + u = u \quad \text{Similarly } u + O = u.$$

(iv) Inverse law

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Every element $v = (a, b) \in V$ has its additive inverse $-v = (-a, -b)$ in V because

$$v + (-v) = (a, b) + (-a, -b) = (a-a, b-b) = (0, 0) = 0$$

$$(-v) + v = (-a, -b) + (a, b) = (-a+a, -b+b) = (0, 0) = 0$$

(v) Commutative law

Let $v_1 = (a, b)$, $v_2 = (c, d) \in V$ then

we prove $v_1 + v_2 = v_2 + v_1$

Now

$$\begin{aligned} v_1 + v_2 &= (a, b) + (c, d) = (a+c, b+d) = (c+a, d+b) \\ &= (c, d) + (a, b) = v_2 + v_1 \end{aligned}$$

Hence $(V, +)$ is an abelian gr.

(b) Scalar multiplication

(i) Let $a \in R$ & $v_1 = (a_1, b_1) \in V$

$$\text{then } av_1 = a(a_1, b_1) = (aa_1, ab_1) \in V$$

(ii) Let $a, b \in R$ & $v_1 = (a_1, b_1) \in V$

$$\text{then we prove } a(bv_1) = (ab)v_1$$

$$\begin{aligned} \text{Now } a(bv_1) &= a(b(a_1, b_1)) = a(ba_1, bb_1) = (a(ba_1), a(bb_1)) \\ &= ((ab)a_1, (ab)b_1) = (ab)(a_1, b_1) = (ab)v_1 \end{aligned}$$

(iii) Let $a, b \in R$ & $v_1 = (a_1, b_1) \in V$ then we prove

$$(a+b)v_1 = av_1 + bv_1$$

$$\text{Now } (a+b)v_1 = (a+b)(a_1, b_1) = ((a+b)a_1, (a+b)b_1) = (aa_1 + ba_1, ab_1 + bb_1)$$

$$+ av_1 + bv_1 = a(a_1, b_1) + b(a_1, b_1) = (aa_1, ab_1) + (ba_1, bb_1) = (aa_1 + ba_1, ab_1 + bb_1)$$

$$\text{So } (a+b)v_1 = av_1 + bv_1$$

Since this condition is not satisfied.

So V is not a vector space over R .

$$(ii) (a, b) + (c, d) = (a, b) \text{ \& } k(a, b) = (ka, kb)$$

$$\text{Sol. Let } V = \{(a, b) \mid a, b \in R\}$$

$$\text{Here } (a, b) + (c, d) = (a, b)$$

$$\& k(a, b) = (ka, kb)$$

First we prove $(V, +)$ is an abelian gr.

(a)

(i) Closure law

Let $u_1 = (a, b)$, $u_2 = (c, d) \in V$ then

$$\begin{aligned} u_1 + u_2 &= (a, b) + (c, d) \\ &= (a, b) \in V \end{aligned}$$

(ii) Associative law

Let $u_1 = (a, b)$, $u_2 = (c, d)$, $u_3 = (e, f) \in V$ then

we prove $u_1 + (u_2 + u_3) = (u_1 + u_2) + u_3$

Now

$$\begin{aligned} u_1 + (u_2 + u_3) &= (a, b) + [(c, d) + (e, f)] \\ &= (a, b) + (c, d) \\ &= (a, b) \end{aligned}$$

$$\begin{aligned} + (u_1 + u_2) + u_3 &= [(a, b) + (c, d)] + (e, f) \\ &= (a, b) + (e, f) \\ &= (a, b) \end{aligned}$$

$$\text{So, } u_1 + (u_2 + u_3) = (u_1 + u_2) + u_3$$

(iii) Identity law

There is no identity element in V

As this condition is not satisfied. So V is not a vector space over $\mathbb{R}$.

$$(iii) \quad (a, b) + (c, d) = (a+c, b+d) \quad + \quad K(a, b) = (K^2 a, K^2 b)$$

Sol. Let $V = \{(a, b) \mid a, b \in \mathbb{R}\}$

$$\text{Here } (a, b) + (c, d) = (a+c, b+d)$$

$$+ \quad K(a, b) = (K^2 a, K^2 b)$$

First we prove that $(V, +)$ is an abelian gr.

(a) (i) Closure law

Let $u_1 = (a, b)$, $u_2 = (c, d) \in V$ then

$$u_1 + u_2 = (a, b) + (c, d) = (a+c, b+d) \in V$$

(ii) Associative law

Let $u_1 = (a, b)$, $u_2 = (c, d)$, $u_3 = (e, f) \in V$

then we prove $v_1 + (v_2 + v_3) = (v_1 + v_2) + v_3$.

Now

$$\begin{aligned} v_1 + (v_2 + v_3) &= (a, b) + [(c, d) + (e, f)] \\ &= (a, b) + (c + e, d + f) \\ &= (a + (c + e), b + (d + f)) \\ &= ((a + c) + e, (b + d) + f) \\ &= (a + c, b + d) + (e, f) \\ &= [(a, b) + (c, d)] + (e, f) \end{aligned}$$

$$\therefore v_1 + (v_2 + v_3) = (v_1 + v_2) + v_3$$

(iii) Identity law

Here $0 = (0, 0)$ is the additive identity in V because for $v = (a, b) \in V$

$$0 + v = (0, 0) + (a, b) = (0 + a, 0 + b) = (a, b) = v$$

$$\& v + 0 = (a, b) + (0, 0) = (a + 0, b + 0) = (a, b) = v$$

(iv) Inverse law

Each element $v = (a, b) \in V$ has its additive inverse $-v = (-a, -b)$ because

$$v + (-v) = (a, b) + (-a, -b) = (a - a, b - b) = (0, 0) = 0$$

$$\& -v + v = (-a, -b) + (a, b) = (-a + a, -b + b) = (0, 0) = 0$$

(v) Commutative law

Let $v_1 = (a, b)$, $v_2 = (c, d) \in V$ then

we prove $v_1 + v_2 = v_2 + v_1$

Now

$$\begin{aligned} v_1 + v_2 &= (a, b) + (c, d) \\ &= (a + c, b + d) \\ &= (c + a, d + b) \\ &= (c, d) + (a, b) \\ &= v_2 + v_1 \end{aligned}$$

Hence $(V, +)$ is an abelian gr.

(b) Scalar multiplication

(i) Let $a \in \mathbb{R}$ & $v_1 = (a_1, b_1) \in V$

$$\text{then } av_1 = a(a_1, b_1) = (a^2 a_1, a^2 b_1) \in V$$

(ii) Let $a, b \in R$ & $u_1 = (a_1, b_1) \in V$

then we prove $a(bu_1) = (ab)u_1$

$$\text{Now } a(bu_1) = a(b(a_1, b_1))$$

$$= a(b^2a_1, b^2b_1)$$

$$= (a^2(b^2a_1), a^2(b^2b_1))$$

$$= ((a^2b^2)a_1, (a^2b^2)b_1)$$

$$= (ab)(a_1, b_1)$$

$$\therefore a(bu_1) = (ab)u_1$$

(iii) Let $a, b \in R$ & $u_1 = (a_1, b_1) \in V$ then we prove

$$(a+b)u_1 = au_1 + bu_1$$

$$\text{Now } (a+b)u_1 = (a+b)(a_1, b_1)$$

$$= ((a+b)a_1, (a+b)b_1)$$

+

$$au_1 + bu_1 = a(a_1, b_1) + b(a_1, b_1)$$

$$= (a^2a_1, a^2b_1) + (b^2a_1, b^2b_1)$$

$$= (a^2a_1 + b^2a_1, a^2b_1 + b^2b_1)$$

$$= ((a^2+b^2)a_1, (a^2+b^2)b_1)$$

$$\text{So } (a+b)u_1 \neq au_1 + bu_1$$

As this condition is not satisfied.

So V is not a vector space over R .

$$(iv) (a, b) + (c, d) = (a+c, b+d) \quad \& \quad K(a, b) = (Ka, 0)$$

Sol.

$$\text{Let } V = \{(a, b) \mid a, b \in R\}$$

$$\text{Here } (a, b) + (c, d) = (a+c, b+d)$$

$$\& \quad K(a, b) = (Ka, 0)$$

First we prove that $(V, +)$ is an abelian gr.

(a) (i) Closure law

$$\text{Let } u_1 = (a, b), u_2 = (c, d) \in V \quad \text{then}$$

$$u_1 + u_2 = (a, b) + (c, d)$$

$$= (a+c, b+d) \in V$$

(ii) Associative law

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Let $u_1 = (a, b)$, $u_2 = (c, d)$, $u_3 = (e, f) \in V$
Then we prove

Now

$$\begin{aligned} u_1 + (u_2 + u_3) &= (u_1 + u_2) + u_3 \\ u_1 + (u_2 + u_3) &= (a, b) + [(c, d) + (e, f)] \\ &= (a, b) + (c+e, d+f) \\ &= (a+(c+e), b+(d+f)) \\ &= ((a+c)+e, (b+d)+f) \\ &= (a+c, b+d) + (e, f) \\ &= [(a, b) + (c, d)] + (e, f) \end{aligned}$$

$$\therefore u_1 + (u_2 + u_3) = (u_1 + u_2) + u_3$$

(iii) Identity law

Here $0 = (0, 0)$ is the additive identity in V
because for $u = (a, b) \in V$

$$0 + u = (0, 0) + (a, b) = (0+a, 0+b) = (a, b) = u$$

$$\& u + 0 = (a, b) + (0, 0) = (a+0, b+0) = (a, b) = u$$

(iv) Inverse law

Each element $u = (a, b) \in V$ has its additive
inverse $-u = (-a, -b)$ in V because

$$u + (-u) = (a, b) + (-a, -b) = (a-a, b-b) = (0, 0) = 0$$

$$\& -u + u = (-a, -b) + (a, b) = (-a+a, -b+b) = (0, 0) = 0$$

(v) Commutative law

Let $u_1 = (a, b)$ & $u_2 = (c, d) \in V$ then we prove

$$u_1 + u_2 = u_2 + u_1$$

$$\begin{aligned} \text{Now } u_1 + u_2 &= (a, b) + (c, d) \\ &= (a+c, b+d) \\ &= (c+a, d+b) \\ &= (c, d) + (a, b) \\ &= u_2 + u_1 \end{aligned}$$

Hence $(V, +)$ is an abelian gr.

(b) Scalar multiplication

(i) Let $a \in R$ & $u_1 = (a_1, b_1) \in V$ then

$$au_1 = a(a_1, b_1) = (aa_1, ab_1) \in V$$

(ii) Let $a, b \in R$ & $u_1 = (a_1, b_1) \in V$. Then we prove

$$a(bu_1) = (ab)u_1$$

Now

$$\begin{aligned} a(bu_1) &= a(b(a_1, b_1)) \\ &= a(ba_1, 0) \\ &= (a(ba_1), 0) \\ &= ((ab)a_1, 0) \\ &= (ab)(a_1, b_1) \end{aligned}$$

$$\therefore a(bu_1) = (ab)u_1$$

(iii) Let $a, b \in R$ & $u_1 = (a_1, b_1) \in V$ then we prove

$$(a+b)u_1 = au_1 + bu_1$$

$$\text{Now } (a+b)u_1 = (a+b)(a_1, b_1)$$

$$\begin{aligned} &= ((a+b)a_1, 0) \\ &= (aa_1 + ba_1, 0) \\ &= (aa_1, 0) + (ba_1, 0) \\ &= a(a_1, b_1) + b(a_1, b_1) \end{aligned}$$

$$\therefore (a+b)u_1 = au_1 + bu_1$$

(iv) Let $a \in R$ & $u_1 = (a_1, b_1), u_2 = (a_2, b_2) \in V$ then

$$\text{we prove } a(u_1 + u_2) = au_1 + au_2$$

$$\begin{aligned} \text{Now } a(u_1 + u_2) &= a((a_1, b_1) + (a_2, b_2)) \\ &= a(a_1 + a_2, b_1 + b_2) \\ &= (a(a_1 + a_2), 0) \\ &= (aa_1 + aa_2, 0) \\ &= (aa_1, 0) + (aa_2, 0) \\ &= a(a_1, b_1) + a(a_2, b_2) \end{aligned}$$

$$\therefore a(u_1 + u_2) = au_1 + au_2$$

(v) Let $1 \in R$ & $u = (a, b) \in V$ then

$$\text{we prove } 1 \cdot u = u$$

$$\text{Now } 1 \cdot u = 1 \cdot (a, b)$$

$$= (1 \cdot a, 0) = (a, 0) \neq (a, b) = u$$

$$\therefore 1 \cdot u \neq u$$

As this condition is not satisfied.

So V is not a vector space over R

Q3 Check whether each of the following is a real vector space.

(i) The set $C[a, b]$ of all Continuous real valued functions defined on $[a, b]$ with the usual operations on functions as

For $f, g \in C[a, b]$ & $\alpha \in \mathbb{R}$

$$(f+g)(x) = f(x) + g(x)$$

$$\& (\alpha f)(x) = \alpha \cdot f(x)$$

Sol.

Let $C[a, b] = \{f : f \text{ is Continuous real valued fn. defined on } [a, b]\}$

then clearly $C[a, b]$ is a subset of the vector space V of all real valued Continuous functions defined on $\mathbb{R}$.

To show that $C[a, b]$ is a vector space over $\mathbb{R}$, we have to show that $C[a, b]$ is a subspace of V

For this

let $f, g \in C[a, b]$

then both f & g are real valued Continuous fns. defined on $[a, b]$.

Then $(f+g)(x) = f(x) + g(x)$ for $x \in [a, b]$

Now $f+g$ being the sum of two Continuous real valued functions defined on $[a, b]$ is also a Continuous real valued function defined on $[a, b]$

So $f+g \in C[a, b]$.

Hence $f, g \in C[a, b] \Rightarrow f+g \in C[a, b]$

Now let $\alpha \in \mathbb{R}$ & $f \in C[a, b]$

then f is a real valued Continuous fn. defined on $[a, b]$.

& $(\alpha f)(x) = \alpha \cdot f(x)$

Clearly scalar multiple of a Continuous real valued function is also a Continuous real valued function defined on $[a, b]$.

So $\alpha \in \mathbb{R}$, $f \in C[a, b] \Rightarrow \alpha f \in C[a, b]$

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Hence $C[a, b]$ is a subspace of the vector space V of all real valued functions defined on $\mathbb{R}$

Hence $C[a, b]$ is a vector space over $\mathbb{R}$

(ii) The set of all functions $f \in C[a, b]$ such that $f(a) = f(b)$

Soln

Let $C'[a, b] = \{f \mid f \in C[a, b] \text{ and } f(a) = f(b)\}$

then clearly $C'[a, b]$ is a subset of $C[a, b]$

To show that $C'[a, b]$ is a vector space over $\mathbb{R}$, we have to show that $C'[a, b]$ is a subspace of the vector space $C[a, b]$.

For this

let $f, g \in C'[a, b]$

then $f, g \in C[a, b]$ s.t.

$$f(a) = f(b) \text{ and } g(a) = g(b)$$

Now $f+g$ being sum of two real valued continuous functions defined on $[a, b]$ is a real valued continuous function defined on $[a, b]$.

Hence $f+g \in C[a, b]$

$$\begin{aligned} \text{Moreover } (f+g)(a) &= f(a) + g(a) \\ &= f(b) + g(b) \end{aligned}$$

$$\Rightarrow (f+g)(a) = (f+g)(b)$$

$$\Rightarrow f+g \in C'[a, b]$$

$$\text{So } f, g \in C'[a, b] \Rightarrow f+g \in C'[a, b]$$

Now

let $\alpha \in \mathbb{R}$ & $f \in C'[a, b]$ then clearly αf is a real valued continuous function defined on $[a, b]$,
 $\Rightarrow \alpha f \in C'[a, b]$

Moreover

$$(af)(a) = af(a)$$

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$$= af(b)$$

$$= f \in C[a, b] \Rightarrow f(a) = f(b)$$

$$(af)(a) = (af)(b)$$

$$\text{Hence } af \in C[a, b]$$

$$\text{So } \alpha \in \mathbb{R}, f \in C[a, b] \Rightarrow \alpha f \in C[a, b]$$

Hence $C[a, b]$ is a subspace of $C[a, b]$

So $C[a, b]$ is a vector space over $\mathbb{R}$



(iii) The set of all solutions of the diff. eq.

$$\frac{d^2y}{dx^2} - 5 \frac{dy}{dx} + 6y = 0$$

Sol.

Let W be the set of all solutions of the given diff. eq. Then W is a subset of the vector space V of all real functions defined on $\mathbb{R}$.

To show that W is a vector space over $\mathbb{R}$, we have to show that W is a subspace of V over $\mathbb{R}$.

For this

let $f, g \in W$ & $a, b \in \mathbb{R}$

then f & g are solns. of diff. eq. $\frac{d^2y}{dx^2} - 5 \frac{dy}{dx} + 6y = 0$

$$\left. \begin{aligned} \text{Then } \frac{d^2f}{dx^2} - 5 \frac{df}{dx} + 6f &= 0 \\ &+ \frac{d^2g}{dx^2} - 5 \frac{dg}{dx} + 6g = 0 \end{aligned} \right\}$$

Now

$$\frac{d^2}{dx^2}(af+bg) - 5 \frac{d}{dx}(af+bg) + 6(af+bg)$$

$$= \frac{d^2}{dx^2}(af) + \frac{d^2}{dx^2}(bg) - 5 \frac{d}{dx}(af) - 5 \frac{d}{dx}(bg) + 6af + 6bg$$

$$= a \frac{d^2f}{dx^2} + b \frac{d^2g}{dx^2} - 5a \frac{df}{dx} - 5b \frac{dg}{dx} + 6af + 6bg$$

$$= a \left(\frac{d^2f}{dx^2} - 5 \frac{df}{dx} + 6f \right) + b \left(\frac{d^2g}{dx^2} - 5 \frac{dg}{dx} + 6g \right)$$

$$= a(0) + b(0)$$

$$= 0$$

So $af+bg$ is a soln. of $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$

Hence $af+bg \in W$

So W is a subspace of V over R

Hence W is a vector space over R .

(18) The set of all 2×2 real matrices of the form

$$\begin{bmatrix} a & 1 \\ 1 & b \end{bmatrix}$$

Sol.

Let $V = \left\{ \begin{bmatrix} a & 1 \\ 1 & b \end{bmatrix} : a, b \in R \right\}$

First we prove that V is an abelian gr. under matrix addition.

This set V has no additive identity

So V is not a gr. under addition.

As this condition is not satisfied.

So V is not a vector space over R .

Q4 Check whether each of the following subsets is a subspace of the indicated vector space:

(i) Q , the set of rational no's. in R

Sol.

Let $a, b \in R$ & $q_1, q_2 \in Q$

Then q_1, q_2 are rational numbers

Since $a, b \in R$ & $q_1, q_2 \in Q$

So $aq_1 + bq_2$ may not be a rational no.

Hence $aq_1 + bq_2 \notin Q$

So $a, b \in R, q_1, q_2 \in Q \Rightarrow aq_1 + bq_2 \notin Q$

Hence Q is not a subspace of R

(11) All 2×2 non singular real matrices in M_{22} .

Sol.

Let V be the set of all non singular real matrices

in M_{22} .

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As additive identity $[0 \ 0]$ of V does not belong to V . So V itself is not a vector space over $\mathbb{R}$. Hence V is not a subspace of M_{22} .

iii) The set $B[a,b]$ of all bounded real functions defined on $[a,b]$ in the space of all real functions defined on $[a,b]$.

Sol.

Here $B[a,b]$ is the set of all bounded real functions defined on $[a,b]$

Let $a, b \in \mathbb{R}$ & $f, g \in B[a,b]$

Then f & g are bounded real functions defined on $[a,b]$ then $af+bg$ is also a bounded real valued function defined on $[a,b]$.

So $af+bg \in B[a,b]$

Hence $a, b \in \mathbb{R}$ & $f, g \in B[a,b] \Rightarrow af+bg \in B[a,b]$

So $B[a,b]$ is the subspace of the vector space of all real functions defined on $[a,b]$.

Q Show that the union of two subspaces of a vector space need not be a subspace. Let X & Y be subspaces of a vector space V . Prove that $X \cup Y$ is a subspace of V if & only if either $X \subset Y$ or $Y \subset X$.

Sol.

Consider the Euclidean space $\mathbb{R}^3$ where

$$\mathbb{R}^3 = \{(x, y, z) : x, y, z \in \mathbb{R}\}$$

$$\text{Let } X = \{(x_1, 0, 0) : x_1 \in \mathbb{R}\}$$

$$\text{ & } Y = \{(0, x_2, 0) : x_2 \in \mathbb{R}\}$$

Then clearly X & Y are subspaces of $\mathbb{R}^3$.

We shall show that $X \cup Y$ is not a subspace of $\mathbb{R}^3$.

Let $x = (x_1, 0, 0)$ & $y = (0, x_2, 0) \in Y$

$\Rightarrow x, y \in X \cup Y$

$$\text{But } x+y = (x_1, 0, 0) + (0, x_2, 0) \\ = (x_1, x_2, 0) \notin X \cup Y$$

As closure property under addition does not hold in $X \cup Y$
So $X \cup Y$ is not a subspace of $\mathbb{R}^3$.

Next

Suppose $X \cup Y$ is a subspace of V & suppose neither $X \subset Y$ nor $Y \subset X$

Then there are elements x & y such that

$x \in X$ but $x \notin Y$ & $y \in Y$ but $y \notin X$.

Now $x, y \in X \cup Y$ & since $X \cup Y$ is a vector space

So $x+y \in X \cup Y$

$\Rightarrow$ either $x+y \in X$ or $x+y \in Y$

Suppose $x+y \in X$

Then $y = (x+y) - x \in X$ (Since X is a vector space)

viz Contradiction

Similarly if $x+y \in Y$

Then $x = (x+y) - y \in Y$ (Since Y is a vector space)

viz again Contradiction

Hence our supposition is wrong.

Hence either $X \subset Y$ or $Y \subset X$

Conversely

Let $X \subset Y$ or $Y \subset X$

$\Rightarrow X \cup Y = Y$ or $X \cup Y = X$

Since X & Y are subspaces of V

Hence $X \cup Y$ is also a subspace of V

Q6 Which of the following are subspaces of $\mathbb{R}^3$?

(i) $W = \{(x, y, z) : x + y + z = 0\}$

Sol.

$$W = \{(x, y, z) : x + y + z = 0\}$$

Let $w_1, w_2 \in W$

$$\Rightarrow w_1 = (x_1, y_1, z_1)$$

$$\& w_2 = (x_2, y_2, z_2) \quad \text{where } x_1 + y_1 + z_1 = 0 \& x_2 + y_2 + z_2 = 0$$

Now let $a, b \in \mathbb{R}$ then

$$\begin{aligned} aw_1 + bw_2 &= a(x_1, y_1, z_1) + b(x_2, y_2, z_2) \\ &= (ax_1, ay_1, az_1) + (bx_2, by_2, bz_2) \\ &= (ax_1 + bx_2, ay_1 + by_2, az_1 + bz_2) \end{aligned}$$

$$\text{Now } aw_1 + bw_2 \in W \text{ if } ax_1 + bx_2 + ay_1 + by_2 + az_1 + bz_2 = 0 \quad \forall a, b \in \mathbb{R}$$

Now

$$\begin{aligned} ax_1 + bx_2 + ay_1 + by_2 + az_1 + bz_2 &= ax_1 + ax_2 + ax_3 + bx_2 + by_2 + bz_2 \\ &= a(x_1 + y_1 + z_1) + b(x_2 + y_2 + z_2) \\ &= a(0) + b(0) \\ &= 0 \end{aligned}$$

$$\text{So } aw_1 + bw_2 \in W$$

$$\text{Hence for } a, b \in \mathbb{R}, w_1, w_2 \in W \Rightarrow aw_1 + bw_2 \in W$$

Hence W is a subspace of $\mathbb{R}^3$.

(ii) $W = \{(x, y, z) : x \geq 0\}$

Sol. Let $w_1, w_2 \in W$

$$\Rightarrow w_1 = (x_1, y_1, z_1)$$

$$\& w_2 = (x_2, y_2, z_2) \quad \text{where } x_1 \geq 0 \& x_2 \geq 0$$

Let $a, b \in \mathbb{R}$ then

$$\begin{aligned} aw_1 + bw_2 &= a(x_1, y_1, z_1) + b(x_2, y_2, z_2) \\ &= (ax_1, ay_1, az_1) + (bx_2, by_2, bz_2) \\ &= (ax_1 + bx_2, ay_1 + by_2, az_1 + bz_2) \end{aligned}$$

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Now $aw_1 + bw_2 \in W$ if $ax_1 + bx_2 \geq 0$

$\forall a, b \in \mathbb{R}$

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As $a, b \in \mathbb{R}$ & $x_1, x_2 \geq 0$

So $ax_1 + bx_2$ may not be ≥ 0

Hence $aw_1 + bw_2 \notin W$ $\forall a, b \in \mathbb{R}$

So W is not a subspace of $\mathbb{R}^2$

(iii) $W = \{ (x, y, z) : x^2 + y^2 + z^2 \leq 1 \}$

Sol:

Let $w_1, w_2 \in W$

$$\Rightarrow w_1 = (x_1, y_1, z_1)$$

$$\& w_2 = (x_2, y_2, z_2) \text{ where } x_1^2 + y_1^2 + z_1^2 \leq 1 \& x_2^2 + y_2^2 + z_2^2 \leq 1$$

Now let $a, b \in \mathbb{R}$ then

$$\begin{aligned} aw_1 + bw_2 &= a(x_1, y_1, z_1) + b(x_2, y_2, z_2) \\ &= (ax_1, ay_1, az_1) + (bx_2, by_2, bz_2) \\ &= (ax_1 + bx_2, ay_1 + by_2, az_1 + bz_2) \end{aligned}$$

Now $aw_1 + bw_2 \in W$ if $(ax_1 + bx_2)^2 + (ay_1 + by_2)^2 + (az_1 + bz_2)^2 \leq 1$

$$\begin{aligned} \text{As } (ax_1 + bx_2)^2 + (ay_1 + by_2)^2 + (az_1 + bz_2)^2 \\ = a^2(x_1^2 + y_1^2 + z_1^2) + b^2(x_2^2 + y_2^2 + z_2^2) + 2ab(x_1x_2 + y_1y_2 + z_1z_2) \end{aligned}$$

If we take

$$a^2 = \frac{1}{x_1^2 + y_1^2 + z_1^2} \& b^2 = \frac{1}{x_2^2 + y_2^2 + z_2^2} \&$$

$a, b, x_1, y_1, z_1, x_2, y_2, z_2$ are all +ve then above expression is $\neq 1$

So $aw_1 + bw_2 \notin W$ $\forall a, b \in \mathbb{R}$

Hence W is not a subspace of $\mathbb{R}^3$

(iv) $W = \{ (x, y, z) : x, y, z \text{ are rationals} \}$

Sol:

Let $w_1, w_2 \in W$ then

$$w_1 = (x_1, y_1, z_1)$$

$$\& w_2 = (x_2, y_2, z_2) \text{ where } x_1, y_1, z_1 \& x_2, y_2, z_2 \text{ are rationals.}$$

Let $a, b \in \mathbb{R}$ then

$$\begin{aligned} aw_1 + bw_2 &= a(x_1, y_1, z_1) + b(x_2, y_2, z_2) \\ &= (ax_1, ay_1, az_1) + (bx_2, by_2, bz_2) \\ &= (ax_1 + bx_2, ay_1 + by_2, az_1 + bz_2) \end{aligned}$$

Now $aw_1 + bw_2 \in W$ if $ax_1 + bx_2, ay_1 + by_2, az_1 + bz_2$ are rational.

Since $a, b \in \mathbb{R}$, so a, b may not be rational. Hence

$ax_1 + bx_2, ay_1 + by_2, az_1 + bz_2$ may not be rational.

So $aw_1 + bw_2 \notin W \quad \forall a, b \in \mathbb{R}$

Hence W is not a subspace of $\mathbb{R}^3$

(V): $W = \{(x, 0, z) : x, z \in \mathbb{R}\}$

Sol:

Let $w_1, w_2 \in W$ then

$$w_1 = (x_1, 0, z_1)$$

$$w_2 = (x_2, 0, z_2) \quad \text{where } x_1, z_1, x_2, z_2 \in \mathbb{R}$$

Let $a, b \in \mathbb{R}$ then

$$\begin{aligned} aw_1 + bw_2 &= a(x_1, 0, z_1) + b(x_2, 0, z_2) \\ &= (ax_1, 0, az_1) + (bx_2, 0, bz_2) \\ &= (ax_1 + bx_2, 0, az_1 + bz_2) \end{aligned}$$

Now $aw_1 + bw_2 \in W$ if $ax_1 + bx_2, az_1 + bz_2 \in \mathbb{R}$.

But as $a, b, x_1, z_1, x_2, z_2 \in \mathbb{R}$.

So $ax_1 + bx_2, az_1 + bz_2 \in \mathbb{R} \quad \forall a, b \in \mathbb{R}$

Hence $aw_1 + bw_2 \in W$

So W is a subspace of $\mathbb{R}^3$

(VI) $W = \{(x, y, z) : y^2 = x + z\}$

Sol: Let $w_1, w_2 \in W$

$$w_1 = (x_1, y_1, z_1)$$

$$w_2 = (x_2, y_2, z_2) \quad \text{where } y_1^2 = x_1 + z_1 \text{ and } y_2^2 = x_2 + z_2$$

Let $a, b \in \mathbb{R}$ then

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$$\begin{aligned}aw_1 + bw_2 &= a(x_1, y_1, z_1) + b(x_2, y_2, z_2) \\&= (ax_1, ay_1, az_1) + (bx_2, by_2, bz_2) \\&= (ax_1 + bx_2, ay_1 + by_2, az_1 + bz_2)\end{aligned}$$

Now $aw_1 + bw_2 \in W$ if $(ay_1 + by_2)^2 = (ax_1 + bx_2)^2 + (az_1 + bz_2)^2$

Now

$$\begin{aligned}(ay_1 + by_2)^2 &= a^2 y_1^2 + b^2 y_2^2 + 2ab y_1 y_2 \\&= a^2 (x_1^2 + z_1^2) + b^2 (x_2^2 + z_2^2) + 2ab y_1 y_2 \\&= (a^2 x_1^2 + b^2 x_2^2) + (a^2 z_1^2 + b^2 z_2^2) + 2ab \sqrt{x_1^2 + z_1^2} \sqrt{x_2^2 + z_2^2} \\&= (a^2 x_1^2 + b^2 x_2^2) + (a^2 z_1^2 + b^2 z_2^2) + 2ab \sqrt{(x_1^2 + z_1^2)(x_2^2 + z_2^2)}\end{aligned}$$

$$\begin{aligned}(ax_1 + bx_2)^2 + (az_1 + bz_2)^2 &= a^2 x_1^2 + b^2 x_2^2 + 2ab x_1 x_2 + a^2 z_1^2 + b^2 z_2^2 + 2ab z_1 z_2 \\&= (a^2 x_1^2 + b^2 x_2^2) + (a^2 z_1^2 + b^2 z_2^2) + 2ab (x_1 x_2 + z_1 z_2)\end{aligned}$$

$$\text{So } (ay_1 + by_2)^2 \neq (ax_1 + bx_2)^2 + (az_1 + bz_2)^2$$

Hence $aw_1 + bw_2 \notin W$

So W is not a subspace of $\mathbb{R}^3$

Q7 Let V be the vector space of all real valued functions defined on $\mathbb{R}$. State which of the following are subspaces of V .

- (i) The set of all even functions.
- (ii) The set of all differentiable functions.
- (iii) The set $W = \{f \mid f(x) = Kf(-x), K \in \mathbb{R} \text{ fixed}\}$
- (iv) The set $W = \{f \in V : \int f(x) dx = 0\}$

Sol.

(i) Here $V = \{f : f \text{ is a real valued fn. defined on } \mathbb{R}\}$

& $W = \{f : f \text{ is an even function}\}$

Let $f, g \in W$

Then both f & g are even functions.

i.e.,

$$f(x) = f(-x) \quad \text{and} \quad g(x) = g(-x)$$

$$f(-x) = f(x)$$

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$$\& g(-x) = g(x)$$

Let $a, b \in \mathbb{R}$ then we prove $af+bg \in W$

Now

$$\begin{aligned} (af+bg)(-x) &= (af)(-x) + (bg)(-x) \\ &= a \cdot f(-x) + b \cdot g(-x) \\ &= af(x) + bg(x) \quad \because f, g \text{ are even fns.} \\ &= (af)(x) + (bg)(x) \\ &= (af+bg)(x) \end{aligned}$$

So $af+bg$ is an even fn. Hence $af+bg \in W$

Hence

$$a, b \in \mathbb{R}, f, g \in W \Rightarrow af+bg \in W$$

So W is a subspace of V .

(ii) $W = \{f: f \text{ is a differentiable function}\}$

Sol. Let $f, g \in W$

then both f & g are differentiable functions. i.e., $f' & g'$ exists.

Now let $a, b \in \mathbb{R}$ then we prove $af+bg \in W$

As

$$\begin{aligned} (af+bg)' &= (af)' + (bg)' \\ &= af' + bg' \end{aligned}$$

Since f', g' exists. Hence $(af+bg)'$ exists.

So $af+bg$ is differentiable. Hence $af+bg \in W$

Hence $a, b \in \mathbb{R} \& f, g \in W \Rightarrow af+bg \in W$

So W is a subspace of V

(iii) $W = \{f: f(x) = kf(-x), k \in \mathbb{R} \text{ fixed}\}$

Sol.

Let $f, g \in W$ then

$$f(x) = kf(-x)$$

$$\& g(x) = kg(-x)$$

where $k \in \mathbb{R}$ is fixed

Let $a, b \in \mathbb{R}$. Then we prove $af + bg \in W$
 Now

$$\begin{aligned}
 (af + bg)(x) &= (af)(x) + (bg)(x) \\
 &= af(x) \\
 &= aKf(-x) + bKg(-x) \\
 &= Ka f(-x) + Kb g(-x) \\
 &= K[(af)(-x) + (bg)(-x)] \\
 &= K[(af + bg)(-x)]
 \end{aligned}$$

$\Rightarrow a, K \in \mathbb{R}$
 $\therefore aK = Ka$

So $af + bg \in W$

Hence $a, b \in \mathbb{R}, f, g \in W \Rightarrow af + bg \in W$

So W is a subspace of V .

(iv) $W = \left\{ f \in V : \int_0^1 f(x) dx = 0 \right\}$

Sol.

Let $f, g \in W$

Hence $\int_0^1 f(x) dx = 0$ and $\int_0^1 g(x) dx = 0$

Now for $a, b \in \mathbb{R}$, we show that $af + bg \in W$

As

$$\begin{aligned}
 \int_0^1 (af + bg)(x) dx &= \int_0^1 [(af)(x) + (bg)(x)] dx \\
 &= \int_0^1 [af(x) + bg(x)] dx \\
 &= \int_0^1 af(x) dx + \int_0^1 bg(x) dx \\
 &= a \int_0^1 f(x) dx + b \int_0^1 g(x) dx \\
 &= a \cdot 0 + b \cdot 0 \\
 &= 0
 \end{aligned}$$

, $\therefore af + bg \in W$

Hence for $a, b \in \mathbb{R}$ & $f, g \in W \Rightarrow af + bg \in W$

So W is a subspace of V .

Q8 Let V be the vector space of all real polynomials of degree $\leq n$ together with the zero polynomial. Determine whether or not W is a subspace of V , where W consists of the zero polynomial and all polynomials

- (i) with integral coefficients & of degree $\leq n$.
- (ii) of degree ≤ 3
- (iii) with only even powers of x & of degree $\leq n$.

Sol.

$$(i) \text{ Here } V = \{a_0 + a_1x + a_2x^2 + \dots + a_nx^n : a_i's \in \mathbb{R}\}$$

$$\& W = \{a_0 + a_1x + a_2x^2 + \dots + a_nx^n : a_i's \in \mathbb{Z}\}$$

Let $w_1, w_2 \in W$ then

$$w_1 = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$$

$$\& w_2 = b_0 + b_1x + b_2x^2 + \dots + b_mx^m \text{ where } a_i, b_i \in \mathbb{Z} \& m \leq n$$

Let $a, b \in \mathbb{R}$ then

$$\begin{aligned} aw_1 + bw_2 &= a(a_0 + a_1x + \dots + a_nx^n) + b(b_0 + b_1x + \dots + b_mx^m) \\ &= aa_0 + aa_1x + \dots + aa_nx^n + bb_0 + bb_1x + \dots + bb_mx^m \\ &= (aa_0 + bb_0) + (aa_1 + bb_1)x + \dots + (aa_n + bb_n)x^n + \dots + aa_nx^n \end{aligned}$$

Since $a, b \in \mathbb{R}$ & $a_i, b_i \in \mathbb{Z}$

s. $aa_i + bb_i$ may not be integers

Hence $aw_1 + bw_2 \notin W \quad \forall a, b \in \mathbb{R}$

s. W is not a subspace of V .

$$(ii) W = \{a_0 + a_1x + \dots + a_nx^n : n \leq 3\}$$

Sol.

Let $w_1, w_2 \in W$ then

$$w_1 = a_0 + a_1x + a_2x^2 + a_3x^3$$

$$\& w_2 = b_0 + b_1x + b_2x^2$$

Let $a, b \in \mathbb{R}$ then

$$aw_1 + bw_2 = a(a_0 + a_1x + a_2x^2 + a_3x^3) + b(b_0 + b_1x + b_2x^2)$$

$$\begin{aligned}
 aw_1 + bw_2 &= aa_0 + aa_1x + aa_2x^2 + aa_3x^3 + bb_0 + bb_1x + bb_2x^2 \\
 &= (aa_0 + bb_0) + (aa_1 + bb_1)x + (aa_2 + bb_2)x^2 + aa_3x^3 \\
 &\text{which is a polynomial of degree } \leq 3
 \end{aligned}$$

Hence $aw_1 + bw_2 \in W$

So W is a subspace of V

(iii) $W = \{ a_0 + a_1x^2 + a_2x^4 + \dots + a_nx^{2n} : n \in \mathbb{Z}^+ \}$
Sol.

Let $w_1, w_2 \in W$ then

$$\begin{aligned}
 w_1 &= a_0 + a_1x^2 + a_2x^4 + \dots + a_nx^{2n} \\
 w_2 &= b_0 + b_1x^2 + b_2x^4 + \dots + b_mx^{2m} \quad \text{where } m < n \\
 &\quad \text{and } m, n \in \mathbb{Z}^+
 \end{aligned}$$

Let $a, b \in \mathbb{R}$ then

$$\begin{aligned}
 aw_1 + bw_2 &= a(a_0 + a_1x^2 + a_2x^4 + \dots + a_nx^{2n}) + b(b_0 + b_1x^2 + b_2x^4 + \dots + b_mx^{2m}) \\
 &= aa_0 + aa_1x^2 + \dots + aa_nx^{2n} + bb_0 + bb_1x^2 + \dots + bb_mx^{2m} \\
 &= (aa_0 + bb_0) + (aa_1 + bb_1)x^2 + \dots + (aa_m + bb_m)x^{2m} + \dots + aa_nx^{2n} \\
 &\text{which is a polynomial with only even powers of } x.
 \end{aligned}$$

Hence $aw_1 + bw_2 \in W$

So W is a subspace of V .

Q9 Express the vector $(2, -5, 3)$ in $\mathbb{R}^3$ as a linear combination of the vectors $(1, -3, 2)$, $(2, -4, -1)$ & $(1, -5, 7)$.

Sol.

$$\begin{aligned}
 \text{Let } (2, -5, 3) &= a(1, -3, 2) + b(2, -4, -1) + c(1, -5, 7) \\
 &= (a, -3a, 2a) + (2b, -4b, -b) + (c, -5c, 7c)
 \end{aligned}$$

$$\text{So } (2, -5, 3) = (a + 2b + c, -3a - 4b - 5c, 2a - b + 7c)$$

$\Rightarrow$

$$a + 2b + c = 2 \quad \text{--- ①}$$

$$-3a - 4b - 5c = -5 \quad \text{--- ②}$$

$$2a - b + 7c = 3 \quad \text{--- ③}$$

Multiplying ① by 3 & adding in ②

$$\begin{aligned} 3a + 6b + 3c &= 6 \\ -3a - 4b - 5c &= -5 \end{aligned}$$

$$2b - 2c = 1$$

$$\text{or } b - c = \frac{1}{2} \quad \text{--- (4)}$$

Now multiplying (1) by 2 & sub. (3) from (1)

$$2a + 4b + 2c = 4$$

$$\begin{aligned} -2a - b + 7c &= 3 \\ \hline 5b - 5c &= 1 \end{aligned}$$

$$\text{or } b - c = \frac{1}{5} \quad \text{--- (5)}$$

From (4) & (5), we cannot find values of b & c

Thus $(2, -5, 3)$ cannot be expressed as a linear combination of $(1, -3, 2)$, $(2, -4, -1)$ & $(1, -5, 7)$

Q1. For what value of k will the vector $(1, -2, k)$ in R^3 be a linear combination of the vectors $(3, 0, -2)$ & $(2, -1, -5)$?

Sol.

$$\begin{aligned} \text{Let } (1, -2, k) &= a(3, 0, -2) + b(2, -1, -5) \\ &= (3a, 0, -2a) + (2b, -b, -5b) \end{aligned}$$

$$\text{or } (1, -2, k) = (3a + 2b, -b, -2a - 5b)$$

$$\Rightarrow \begin{aligned} 3a + 2b &= 1 & \text{--- (1)} \\ -b &= -2 & \text{--- (2)} \\ -2a - 5b &= k & \text{--- (3)} \end{aligned}$$

$$\text{from (2) } b = 2$$

$$\text{Put in (1)}$$

$$3a + 2(2) = 1$$

$$3a + 4 = 1$$

$$3a = -3$$

$$a = -1$$

Putting values of a & b in ③

$$-2(-1) - 5(2) = K$$

$$2 - 10 = K$$

$$-8 = K$$

or $K = -8$

So for $K = -8$, the vector $(1, -2, K)$ is a linear combination of $(3, 0, -2)$ & $(2, -1, -5)$.

Q11 Let U & W be the subspaces of $\mathbb{R}^3$ defined by
 $U = \{(x, y, z) : x = y = z\}$
 & $W = \{(0, y, z) : y, z \in \mathbb{R}\}$

Show that

$$\mathbb{R}^3 = U \oplus W$$

Sol.

To show that $\mathbb{R}^3 = U \oplus W$, we have to prove

that $\left. \begin{aligned} \mathbb{R}^3 &= U + W \\ &\text{& } U \cap W = \{0\} \end{aligned} \right\}$

Here

$$U = \{(x, y, z) : x = y = z\}$$

$$\text{& } W = \{(0, y, z) : y, z \in \mathbb{R}\}$$

Let

$$(x, y, z) \in \mathbb{R}^3$$

$$\text{then } (x, y, z) = (x, x, x) + (0, y-x, z-x) \in U + W$$

$$\text{So } (x, y, z) \in U + W$$

$$\text{Hence } \mathbb{R}^3 \subseteq U + W \quad \text{————— ①}$$

Conversely

$$\text{let } u \in U \text{ & } w \in W \text{ then } u + w \in U + W$$

$$\Rightarrow u = (x, x, x) \text{ & } w = (0, y, z)$$

$$\text{then } u + w = (x, x, x) + (0, y, z)$$

$$= (x, x+y, x+z) \in \mathbb{R}^3$$

So $U+W \in \mathbb{R}^3$

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Hence $U+W \subseteq \mathbb{R}^3$ ———— ②

From ① & ②

$$\mathbb{R}^3 = U+W$$

Now we want to show $U \cap W = \{0\}$

Let $\alpha \in U \cap W$

$$\Rightarrow \alpha \in U \text{ \& \> } \alpha \in W$$

$$\Rightarrow \alpha = (x, x, x) \text{ \& \> } \alpha = (0, y, z)$$

$$\Rightarrow (x, x, x) = (0, y, z)$$

$$\text{So } x = 0, \quad x = y, \quad x = z$$

$$\Rightarrow x = y = z = 0$$

$$\text{So } \alpha = (0, 0, 0)$$

$$\text{Hence } U \cap W = \{0\}$$

So

$$\mathbb{R}^3 = U \oplus W$$

Q12 Show that each of the following sets of vectors generate $\mathbb{R}^3$

(i) $\{(1, 2, 3), (0, 1, 2), (0, 0, 1)\}$

(ii) $\{(1, 1, 1), (0, 1, 1), (0, 1, -1)\}$

Soln-

(i) Given set is $\{(1, 2, 3), (0, 1, 2), (0, 0, 1)\}$

Let $(x, y, z) \in \mathbb{R}^3$ & suppose

$$(x, y, z) = a(1, 2, 3) + b(0, 1, 2) + c(0, 0, 1)$$

$$= (a, 2a, 3a) + (0, b, 2b) + (0, 0, c)$$

$$\text{So } (x, y, z) = (a, 2a+b, 3a+2b+c)$$

$$\Rightarrow a = x \text{ ————— ①}$$

$$2a+b = y \text{ ————— ②}$$

$$3a+2b+c = z \text{ ————— ③}$$

from ①

$$\boxed{a = x}$$

Put in ②

$$2x + b = y$$

$$b = y - 2x$$

Put values of a & b in ③

$$3x + 2(y - 2x) + c = z$$

$$3x + 2y - 4x + c = z$$

$$\text{or } -x + 2y + c = z$$

$$\text{or } c = x - 2y + z$$

So

$$(x, y, z) = x(1, 2, 3) + (y - 2x)(0, 1, 2) + (x - 2y + z)(0, 0, 1)$$

Hence given vectors generate $\mathbb{R}^3$.(ii) Given set is $\{(1, 1, 1), (0, 1, 1), (0, 1, -1)\}$ Let $(x, y, z) \in \mathbb{R}^3$ & suppose

$$(x, y, z) = a(1, 1, 1) + b(0, 1, 1) + c(0, 1, -1)$$

$$= (a, a, a) + (0, b, b) + (0, c, -c)$$

$$\text{u} (x, y, z) = (a, a + b + c, a + b - c)$$

 $\Rightarrow$

$$a = x \quad \text{--- ①}$$

$$a + b + c = y \quad \text{--- ②}$$

$$a + b - c = z \quad \text{--- ③}$$

$$\text{from ① } a = x$$

Add ② & ③

$$2a + 2b = y + z$$

$$2x + 2b = y + z$$

$$\text{or } 2b = y + z - 2x$$

$$\text{or } b = \frac{y + z - 2x}{2}$$

Put values of a & b in ②

$$x + \frac{y + z - 2x}{2} + c = y$$

$$c = y - x - \frac{y+z-2x}{2}$$

$$= \frac{2y-2x-y-z+2x}{2}$$

$$c = \frac{y-z}{2}$$

So

$$(x, y, z) = x(1, 1, 1) + \left(\frac{y+z-2x}{2}\right)(0, 1, 1) + \left(\frac{y-z}{2}\right)(0, 1, -1)$$

Hence given vectors generate $\mathbb{R}^3$.

Q13 Determine whether the set $S = \{(1, 1, 2), (1, 0, 1), (2, 1, 3)\}$ spans $\mathbb{R}^3$

Sol: Given set is $S = \{(1, 1, 2), (1, 0, 1), (2, 1, 3)\}$

Let $(x, y, z) \in \mathbb{R}^3$ & suppose

$$\begin{aligned}(x, y, z) &= a(1, 1, 2) + b(1, 0, 1) + c(2, 1, 3) \\ &= (a, a, 2a) + (b, 0, b) + (2c, c, 3c)\end{aligned}$$

$$\therefore (x, y, z) = (a+b+2c, a+c, 2a+b+3c)$$

$\Rightarrow$

$$a + b + 2c = x \quad \text{--- (1)}$$

$$a + c = y \quad \text{--- (2)}$$

$$2a + b + 3c = z \quad \text{--- (3)}$$

Sol: (2) from (1)

$$-a - c = x - z$$

$$\text{or } a + c = z - x \quad \text{--- (4)}$$

$$\& \quad a + c = y \quad \text{--- (2)}$$

The eqs. (2) & (4) cannot be solved for a & c

Hence we cannot find values of a, b & c .

So the set $S = \{(1, 1, 2), (1, 0, 1), (2, 1, 3)\}$

does not span $\mathbb{R}^3$

Q14 Show that the yz -plane

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$W = \{(0, y, z) : y, z \in \mathbb{R}\}$ is spanned by

(i) $(0, 1, 1)$ and $(0, 2, -1)$

(ii) $(0, 1, 2)$, $(0, 2, 3)$ & $(0, 3, 1)$

Sol.

(i) Let $W = \{(0, y, z) : y, z \in \mathbb{R}\}$

Suppose

$$(0, y, z) \in W$$

$$(0, y, z) = a(0, 1, 1) + b(0, 2, -1)$$

$$= (0, a, a) + (0, 2b, -b)$$

$$\therefore (0, y, z) = (0, a+2b, a-b)$$

$$\Rightarrow a+2b = y \quad \text{--- (1)}$$

$$a-b = z \quad \text{--- (2)}$$

Solnt. (2) from (1)

$$3b = y-z$$

$$b = \frac{y-z}{3}$$

Put value in (2)

$$a - \frac{y-z}{3} = z$$

$$\text{or } a = \frac{y-z}{3} + z$$

$$= \frac{y-z+3z}{3}$$

$$\therefore a = \frac{y+2z}{3}$$

S.

$$(0, y, z) = \left(\frac{y+2z}{3}\right)(0, 1, 1) + \left(\frac{y-z}{3}\right)(0, 2, -1)$$

Hence yz -plane is spanned by $(0, 1, 1)$ & $(0, 2, -1)$

(ii) Sol. Given vectors are $(0, 1, 2)$, $(0, 2, 3)$ & $(0, 3, 1)$

Let $(0, y, z) \in W$ & suppose

$$(0, y, z) = a(0, 1, 2) + b(0, 2, 3) + c(0, 3, 1)$$

$$= (0, a, 2a) + (0, 2b, 3b) + (0, 3c, c)$$

$$\text{So } (0, y, z) = (0, a + 2b + 3c, 2a + 3b + c)$$

$\Rightarrow$

$$a + 2b + 3c = y \quad \text{--- (1)}$$

$$2a + 3b + c = z \quad \text{--- (2)}$$

Put $a = 0$

$$\text{So } 2b + 3c = y \quad \text{--- (3)}$$

$$3b + c = z \quad \text{--- (4)}$$

Multiplying (4) by 3 & subst. from (3)

$$2b + 3c = y \quad \text{--- (3)}$$

$$-9b + 3c = -3z \quad \text{--- (5)}$$

$$-7b = y - 3z$$

$$b = \frac{3z - y}{7}$$

Put in (3)

$$2\left(\frac{3z - y}{7}\right) + 3c = y$$

$$\frac{6z - 2y}{7} + 3c = y$$

$$3c = y - \frac{6z - 2y}{7}$$

$$= \frac{7y - 6z + 2y}{7}$$

$$3c = \frac{9y - 6z}{7}$$

$$c = \frac{3y - 2z}{7}$$

$$\text{Hence } (0, y, z) = 0(0, 1, 2) + \left(\frac{3z - y}{7}\right)(0, 2, 3) + \left(\frac{3y - 2z}{7}\right)(0, 3, 1)$$

So yz -plane is spanned by $(0, 1, 2), (0, 2, 3)$ & $(0, 3, 1)$

Q15 Find an eq. (or equations) of the subspace W of $\mathbb{R}^3$ generated by each of following sets of vectors.

(i) $\{(1, -3, 5), (-2, 6, -10)\}$

(ii) $\{(1, -3, 2), (-2, 0, 3)\}$

(iii) $\{(1, -2, 1), (-2, 0, 3), (3, 2, -2)\}$

(i) $\{(1, -3, 5), (-2, 6, -10)\}$

Sol. Since W is spanned by the vectors $(1, -3, 5)$, $(-2, 6, -10)$.

So each vector $(x, y, z) \in W$ is a linear combination of these vectors. i.e., there exist scalars a, b s.t.

$$\begin{aligned}(x, y, z) &= a(1, -3, 5) + b(-2, 6, -10) \\ &= (a, -3a, 5a) + (-2b, 6b, -10b)\end{aligned}$$

$$\text{or } (x, y, z) = (a - 2b, -3a + 6b, 5a - 10b)$$

$$\Rightarrow \left. \begin{aligned} a - 2b &= x \\ -3a + 6b &= y \\ 5a - 10b &= z \end{aligned} \right\} \text{----- ①}$$

We reduce the augmented matrix of system ① to echelon form as:

$$A_b = \begin{bmatrix} 1 & -2 & x \\ -3 & 6 & y \\ 5 & -10 & z \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & -2 & x \\ 0 & 0 & y+3x \\ 0 & 0 & z-5x \end{bmatrix} \quad \begin{array}{l} R_2 + 3R_1 \\ R_3 - 5R_1 \end{array}$$

The system ① is consistent if
 $\text{rank } A = \text{rank } A_b$

$$\Rightarrow y + 3x = 0 \quad \& \quad z - 5x = 0$$

$$\text{or } \left. \begin{aligned} x &= t \\ y &= -3t \\ z &= 5t \end{aligned} \right\} \quad t \in \mathbb{R}$$

These are the req. eqs. of the subspace W of $\mathbb{R}^3$

(ii) $\{(1, -3, 2), (-2, 0, 3)\}$

Sol. Since W is spanned by the vectors $(1, -3, 2)$ & $(-2, 0, 3)$. So each vector $(x, y, z) \in W$ is a

Linear Combination of these vectors.

i.e., there exist scalars $a, b \in \mathbb{R}$ s.t.

$$\begin{aligned}(x, y, z) &= a(1, -3, 2) + b(-2, 0, 3) \\ &= (a, -3a, 2a) + (-2b, 0, 3b)\end{aligned}$$

$$a(x, y, z) = (a - 2b, -3a, 2a + 3b)$$

$\Rightarrow$

$$\left. \begin{aligned} a - 2b &= x \\ -3a &= y \\ 2a + 3b &= z \end{aligned} \right\} \text{--- ①}$$

We reduce the augmented matrix of system ① to echelon form as follows:

$$A_b = \begin{bmatrix} 1 & -2 & x \\ -3 & 0 & y \\ 2 & 3 & z \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & -2 & x \\ 0 & -6 & y+3x \\ 0 & 7 & z-2x \end{bmatrix}$$

$R_2 + 3R_1$

$R_3 - 2R_1$

$$\sim \begin{bmatrix} 1 & -2 & x \\ 0 & 1 & -\frac{1}{6}(y+3x) \\ 0 & 1 & \frac{1}{7}(z-2x) \end{bmatrix}$$

$-\frac{1}{6}R_2, \frac{1}{7}R_3$

$$\sim \begin{bmatrix} 1 & -2 & x \\ 0 & 1 & -\frac{1}{6}(y+3x) \\ 0 & 0 & \frac{1}{7}(z-2x) + \frac{1}{6}(y+3x) \end{bmatrix}$$

$R_3 - R_2$

The system ① is consistent if the
rank $A = \text{rank } A_b$

$$\Rightarrow \frac{1}{7}(z-2x) + \frac{1}{6}(y+3x) = 0$$

$$\text{or } 6(z-2x) + 7(y+3x) = 0$$

$$6z - 12x + 7y + 21x = 0$$

$$9x + 7y + 6z = 0$$

which is the req. eq. of the subspace W of $\mathbb{R}^3$.

(iii) $\{(1, -2, 1), (-2, 0, 3), (3, -2, -2)\}$ 550

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Solⁿ. Since W is spanned by the vectors $(1, -2, 1)$, $(-2, 0, 3)$ & $(3, -2, -2)$. So each vector $(x, y, z) \in W$ is a linear combination of these vectors.

i.e., there exist scalars $a, b, c \in \mathbb{R}$ s.t.

$$\begin{aligned}(x, y, z) &= a(1, -2, 1) + b(-2, 0, 3) + c(3, -2, -2) \\ &= (a, -2a, a) + (-2b, 0, 3b) + (3c, -2c, -2c)\end{aligned}$$

$$\text{or } (x, y, z) = (a - 2b + 3c, -2a - 2c, a + 3b - 2c)$$

$$\Rightarrow \begin{cases} a - 2b + 3c = x \\ -2a - 2c = y \\ a + 3b - 2c = z \end{cases} \quad \text{--- (1)}$$

We reduce the augmented matrix of this system to echelon form as:

$$A_b = \begin{bmatrix} 1 & -2 & 3 & x \\ -2 & 0 & -2 & y \\ 1 & 3 & -2 & z \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & -2 & 3 & x \\ 0 & -4 & 4 & y+2x \\ 0 & 5 & -5 & z-x \end{bmatrix} \quad \begin{array}{l} R_2 + 2R_1 \\ R_3 - R_1 \end{array}$$

$$\sim \begin{bmatrix} 1 & -2 & 3 & x \\ 0 & 1 & -1 & -\frac{1}{4}(y+2x) \\ 0 & 1 & -1 & \frac{1}{5}(z-x) \end{bmatrix} \quad -\frac{1}{4}R_2, \frac{1}{5}R_3$$

$$\sim \begin{bmatrix} 1 & -2 & 3 & x \\ 0 & 1 & -1 & -\frac{1}{4}(y+2x) \\ 0 & 0 & 0 & \frac{1}{5}(z-x) + \frac{1}{4}(y+2x) \end{bmatrix} \quad R_3 - R_2$$

The system (1) is consistent if

$$\text{rank } A = \text{rank } A_b$$

$$\Rightarrow \frac{1}{5}(z-x) + \frac{1}{4}(y+2x) = 0$$

$$\Rightarrow 4(z-x) + 5(y+2x) = 0 \quad 551$$

$$\text{or } 4z - 4x + 5y + 10x = 0$$

$$\text{or } 6x + 5y + 4z = 0$$

Which is the req. eq. of subspace W of $\mathbb{R}^3$

Q16 Show that the Complex no's. $2+3i$ & $1-2i$ generate the vector space $\mathbb{C}$ over $\mathbb{R}$.

Sol. Here $\mathbb{C} = \{x+iy : x, y \in \mathbb{R}\}$

Any vector of $\mathbb{C}$ has the form $x+iy$; $x, y \in \mathbb{R}$

Suppose

$$x+iy = a(2+3i) + b(1-2i)$$

$$= 2a + 3ai + b - 2bi$$

$$\text{or } x+iy = (2a+b) + i(3a-2b)$$

$\Rightarrow$

$$2a+b = x \quad \text{--- (1)}$$

$$3a-2b = y \quad \text{--- (2)}$$

Multiplying (1) by 2 & adding in (2)

$$4a+2b = 2x \quad \text{--- (1)}$$

$$3a-2b = y \quad \text{--- (2)}$$

$$7a = 2x+y$$

$$a = \frac{2x+y}{7}$$

Put in (1)

$$2\left(\frac{2x+y}{7}\right) + b = x$$

$$\frac{4x+2y}{7} + b = x$$

$$b = x - \frac{4x+2y}{7}$$

$$= \frac{7x-4x-2y}{7}$$

$$b = \frac{3x-2y}{7}$$



Hence

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$$x+iy = \left(\frac{2x+y}{7}\right)(2+3i) + \left(\frac{3x-2y}{7}\right)(1-2i)$$

So given vectors $2+3i$ & $1-2i$ generate C over R .

Q17 Let S & T be subsets of a vector space V .
Show that

(i) $\langle S \rangle \cup \langle T \rangle \subseteq \langle S \cup T \rangle$

(ii) $\langle S \cap T \rangle \subseteq \langle S \rangle \cap \langle T \rangle$

Give an example to show that equality need not hold in either case.

Sol.

(i) Let $S = \{u_1, u_2, \dots, u_r\}$ &

$$T = \{v_1, v_2, \dots, v_t\}$$

$$\Rightarrow S \cup T = \{u_1, u_2, \dots, u_r, v_1, v_2, \dots, v_t\}$$

We want to show $\langle S \rangle \cup \langle T \rangle \subseteq \langle S \cup T \rangle$

Let $v \in \langle S \rangle \cup \langle T \rangle$

$$\Rightarrow v \in \langle S \rangle \text{ or } v \in \langle T \rangle$$

Let $v \in \langle S \rangle$ then v is a linear combination of vectors of S

$$\text{i.e., } v = k_1 u_1 + k_2 u_2 + \dots + k_r u_r$$

We can also write v as

$$v = k_1 u_1 + k_2 u_2 + \dots + k_r u_r + 0v_1 + 0v_2 + \dots + 0v_t$$

which shows that v is a linear combination of vectors of $S \cup T$

$$\Rightarrow v \in \langle S \cup T \rangle$$

$$\text{S. } v \in \langle S \rangle \Rightarrow v \in \langle S \cup T \rangle$$

$$\text{Similarly } v \in \langle T \rangle \Rightarrow v \in \langle S \cup T \rangle$$

Hence

$$\langle S \rangle \cup \langle T \rangle \subseteq \langle S \cup T \rangle$$

Now we give an example to show that equality does not hold in above result.

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in $\mathbb{R}^2$, let $S = \{(1,0)\}$, $T = \{(0,1)\}$ then

$$\langle S \rangle = \{k(1,0) : k \in \mathbb{R}\} = x\text{-axis}$$

$$\& \langle T \rangle = \{l(0,1) : l \in \mathbb{R}\} = y\text{-axis}$$

Therefore

$$\langle S \rangle \cup \langle T \rangle = \{k(1,0) : k \in \mathbb{R}\} \cup \{l(0,1) : l \in \mathbb{R}\} \neq \mathbb{R}^2$$

Now

$$S \cup T = \{(1,0), (0,1)\} \text{ and}$$

Any vector $(x,y) \in \mathbb{R}^2$ is a linear combination of $(1,0)$ & $(0,1)$, because

$$(x,y) = x(1,0) + y(0,1)$$

Therefore

$$\langle S \cup T \rangle = \mathbb{R}^2$$

This shows that $\langle S \rangle \cup \langle T \rangle \neq \langle S \cup T \rangle$

$$(ii) \quad \langle S \cap T \rangle \subset \langle S \rangle \cap \langle T \rangle$$

Sol.

First we prove that if

$$S \subset T \text{ then } \langle S \rangle \subset \langle T \rangle$$

$$\text{Let } S = \{v_1, v_2, \dots, v_h\}$$

$$\& T = \{v_1, v_2, \dots, v_h, v_{h+1}, \dots, v_n\}$$

then obviously $S \subset T$

We want to show that $\langle S \rangle \subset \langle T \rangle$

$$\text{Let } v \in \langle S \rangle$$

then v is a linear combination of vectors of S .

$$\text{i.e., } v = a_1 v_1 + a_2 v_2 + \dots + a_h v_h$$

Now v can also be written as

$$v = a_1 v_1 + a_2 v_2 + \dots + a_h v_h + 0 v_{h+1} + \dots + 0 v_n$$

Which shows v is a linear combination of vectors of T . Hence $v \in \langle T \rangle$

$$\text{So } \langle S \rangle \subset \langle T \rangle$$

Now as $S \cap T \subset S$ & $S \cap T \subset T$

$$\text{So } \langle S \cap T \rangle \subset \langle S \rangle \text{ & } \langle S \cap T \rangle \subset \langle T \rangle$$

$$\Rightarrow \langle S \cap T \rangle \subset \langle S \rangle \cap \langle T \rangle$$

Now we will give an example to show that equality does not hold in above result.

Example

In $\mathbb{R}^2$, let $S = \{(0,0), (1,0)\}$ & $T = \{(0,0), (0,3)\}$

then $S \cap T = \{(0,0)\}$

$$\begin{aligned} \text{So } \langle S \cap T \rangle &= \{k(0,0) : k \in \mathbb{R}\} \\ &= \{(0,0)\} \quad \text{--- (i)} \end{aligned}$$

Now

$$\begin{aligned} \langle S \rangle &= \{a(0,0) + b(1,0) : a, b \in \mathbb{R}\} \\ &= \{(0, b) : b \in \mathbb{R}\} \end{aligned}$$

&

$$\begin{aligned} \langle T \rangle &= \{p(0,0) + q(0,3) : p, q \in \mathbb{R}\} \\ &= \{(0, 3q) : q \in \mathbb{R}\} \\ &= \{(0, c) : c \in \mathbb{R}\} \end{aligned}$$

Now

$$\begin{aligned} \langle S \rangle \cap \langle T \rangle &= \{(0, b) : b \in \mathbb{R}\} \cap \{(0, c) : c \in \mathbb{R}\} \\ &= \{(0, y) : y \in \mathbb{R}\} \quad \text{--- (ii)} \end{aligned}$$

From (i) & (ii)

$$\langle S \cap T \rangle \neq \langle S \rangle \cap \langle T \rangle$$